Indian Market Disclosures in One Place

A free data platform and web application for India's public stock market disclosures

Project review report

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Date: 7 November 2026

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1. Introduction

Every day, Indian stock exchanges and regulators publish a lot of useful information for free. They publish who made large trades, who owns each company, how much foreign investors bought and sold, what each mutual fund holds, and what profit each company made.

All of this is public, and anybody can read it. But reading it is not easy. The information is spread over many websites and in many file formats. Some of it is removed the very next day.

This project collects all of that information every evening, keeps it safely, cleans it, and shows it in one simple website.

What I built

I built three things that work together:

- **A collector.** It runs by itself every evening. It downloads the day's files and keeps the original copies.
- **A database.** It turns those files into clean tables and links everything to the right company.
- **A web application.** It lets anyone look up a stock, a fund, an investor or a day's trades in one place.

Why I chose this project

I started this project because I wanted to understand whether large investors' public trades tell us anything about share prices. To answer that, I first needed the data. I then found that collecting it was the hardest part of the whole job.

So the data platform became the main work, and the question about share prices became one of the things it can be used for.

What this project does not do

This project is not a trading system, and three things make that clear:

- It does not place any orders, and it holds no money.
- It does not tell anyone to buy or sell a share. It gives no scores or ratings.
- It only shows what was officially disclosed, and it says where each number came from.

2. The problem

The data is free, but there is no single proper place to get all of it.

I looked for one free place that had all of this data together, going back many years. I could not find one. Paid terminals exist, and free websites each cover some parts, but I had to go to many different places to build the full picture of even one company.

The problem has six parts.

2.1 The data is scattered

To understand one company, you may need to visit all of these:

What you want to know	Where it is published
Who made a large trade today	NSE, as a daily list
What the share price and delivery were	NSE, in daily files
Who owns the company	Quarterly filings on NSE and BSE
What profit it made	Quarterly filings on NSE
Did the promoters buy or sell	NSE insider trading filings
How much foreign investors bought overall	NSDL
Which mutual funds hold it	Each fund house's own website

No single page joins these together. For mutual funds it is especially hard, because each fund house publishes its own files on its own website, in its own layout.

2.2 Some data disappears

This surprised me the most. Some pages only show the current day, and the next day the old information is gone.

Data	How long it stays available
NSE bulk and block deals	Current day only. The history service returned an error when I tried it
FII and DII cash figures	Current day only
Insider trades	A rolling window of about four months

So if nobody saves this data on the day, it may be lost for good. My own project lost 33 sessions of bulk deals from 2026 this way, and those days cannot be recovered.

2.3 Every source has a different format

The data comes as CSV files, JSON from websites, XBRL files for financial results, and Excel files from fund houses. Each one has to be read in its own way, and a format can change without notice.

2.4 The same thing has many names

This is a bigger problem than it sounds.

- **Companies change names and symbols.** CADILAHC became ZYDUSLIFE, for example. A simple search on the new symbol misses the old history.
- **Investors are spelled in many ways.** One institution can appear under several spellings in the trade lists. I found about 29,877 different spellings, which group into about 27,998 participants.
- **Share splits change the price.** After a 1-for-2 split, the price falls by half on paper, though nobody has lost money. Without correcting for this, charts show false crashes.

2.5 The raw lists can mislead

Much of the bulk deal list is not long-term investors at all. In the 6 November session, 59% of the rows were same-day round trips, where one party bought and sold the same share on the same day. A person reading the raw list could easily believe big investors were buying, when these were mostly high-frequency traders.

2.6 It is hard for a new person

A student or a new investor does not know where to start. Even a person who knows the sources has to download files, clean them, and match them by hand. That takes many days before they can ask a single question.

The goal

So the goal of this project became simple:

> Bring India's free market disclosures into one place, keep them safe, make them correct, and make them easy to read.

3. How I solved it

The solution has seven simple steps. Each one fixes a part of the problem above.



The seven steps, from a public file to a page on the website.

Step 1: Collect every evening

A scheduled job starts by itself at 18:30 and again at 21:00, every day, because some files are published late. Nobody has to press a button. This fixes the problem of disappearing data, because each day's file is saved on the day.

Step 2: Keep the original file

Each file is saved exactly as it was downloaded, compressed, and given a unique fingerprint called a hash. These files are never changed. If I later find a mistake in how I read a file, I can read it again from the original.

The archive now holds more than 1,92,000 files.

Step 3: Read and clean

A small reader for each format turns the file into a clean table. This fixes the problem of many formats, because everything after this step looks the same.

Step 4: Match the names

Each company gets one record, based on its ISIN, which is a code that stays with the share. The system also remembers which symbol was valid on which dates. So old and new names lead to the same company.

Investor spellings are grouped in the same way, so one institution is shown as one participant.

Step 5: Add labels

The system labels each participant as a mutual fund, a foreign institution, an individual, a company and so on. It also finds high-frequency traders from what they do, not from their names: if a party buys and sells the same share on the same day again and again, it gets that label.

On the website, these round trips are folded away by default, and one click shows them. This fixes the misleading lists.

Step 6: Check everything

I do not assume the data is right. I test it:

- **1280 automated tests** check the code and the data.
- I compared my share prices with a separate source across 2,651 companies. This found 226 missed splits and bonuses, which I corrected, and 203 bad one-day prices, which I removed.
- A health check runs after every collection and reports any source that has gone stale.

Step 7: Show it simply

The website shows everything in one place, with plain explanations next to the numbers. A blank is left blank. The website never fills a gap with a guess.

Three rules I followed

- **Show facts, not advice.** The website does not say buy or sell. This is also the safe side of SEBI's rules for anyone who is not a registered adviser.
- **Respect the websites.** I download public files at a slow rate. Where a fund house blocks automated access, I leave it out, and the website says which houses are covered.
- **Be honest about gaps.** If data is missing, the system records this. It does not hide it.

4. The data sources and the database

4.1 Where the data comes from

Source	What I collect
NSE	Bulk and block deals, daily prices, delivery figures, F&O contracts, positions by participant type, FII and DII cash, index values and members, corporate actions, insider trades, quarterly results, shareholding filings, announcements, board meetings and promoter pledges
BSE	Shareholding filings
NSDL	Foreign portfolio investor flows
28 mutual fund houses	Month-end portfolios of their schemes, from each house's own website
Earlier work	The history I collected in the first version of this project, including records the exchanges no longer serve

4.2 How much data there is

These are the numbers from the system's own coverage report.

Data	Rows	From	To
Daily share prices	78,80,786	Jan 2005	Nov 2026
F&O contracts	18,11,60,973	Jan 2005	Nov 2026
Delivery figures	87,69,965	Jan 2006	Nov 2026
Bulk and block deals	2,45,657	Jan 2006	Nov 2026
Foreign investor flows	51,756	Dec 1998	Nov 2026
Index total return values	7,893	Jan 1995	Nov 2026
Quarterly results	76,920	Mar 2005	Sep 2026
Shareholding rows	23,38,249	Mar 2008	Oct 2026
Named shareholders above 1%	16,72,761	Mar 2008	Oct 2026
Mutual fund holdings	27,86,334	Sep 2012	Oct 2026
Insider trades, from earlier work	2,83,281	2016	mid 2026
Insider trades, collected directly	6,710	May 2024	Nov 2026

In total, the project holds about 19 GB of data on disk.

4.3 What each kind of data means

Bulk deal. When one client trades more than 0.5% of a company's shares in a day, the exchange must publish it, with the client's name.

Block deal. A large trade made in a special window of the exchange.

Delivery. The share of traded quantity that really changed hands, and was not just bought and sold within the day.

FII and DII. Foreign and domestic institutions. Their daily totals show which way big money moved.

F&O positions. How many futures contracts each group of traders holds. This is different from buying shares, and the website keeps the two apart.

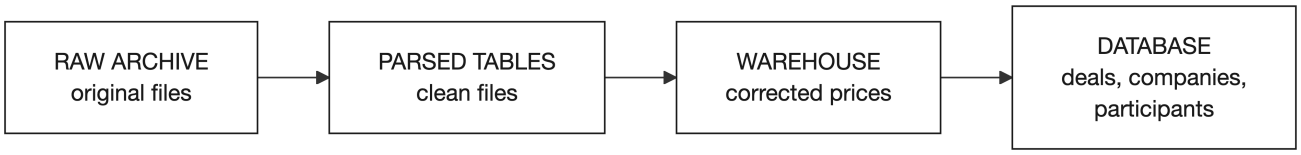
Shareholding pattern. Every quarter, each company files who owns it: promoters, foreign investors, mutual funds and the public.

Insider trades. When promoters or senior staff trade their own company's shares, they must disclose it.

Mutual fund portfolio. Every month, each fund house publishes what each of its schemes holds.

4.4 How the data is stored

The storage has four layers. Data moves one way, from top to bottom.



The four storage layers.

Layer	What it holds	Size
Raw archive	Original downloaded files	3.8 GB
Parsed tables	Clean Parquet files	417 MB
Warehouse	Price tables and comparison groups	5.5 GB
Databases	Linked tables and the project's records	1.2 GB

Only the raw archive is precious. Everything below it can be rebuilt from it.

4.5 The main database tables

Table	Rows	What it holds
`security_master`	3,421	One record per company
`symbol_history`	3,735	Which symbol was valid when
`participant_master`	27,998	One record per investor
`participant_aliases`	29,877	Each spelling, and who it belongs to
`institutional_deals_raw`	2,45,657	Deals exactly as filed
`institutional_deals_clean`	2,45,657	Deals with labels and flags
`participant_oi`	15,819	Daily F&O positions by group
`promoter_entities`	1,05,357	Promoters named in company filings

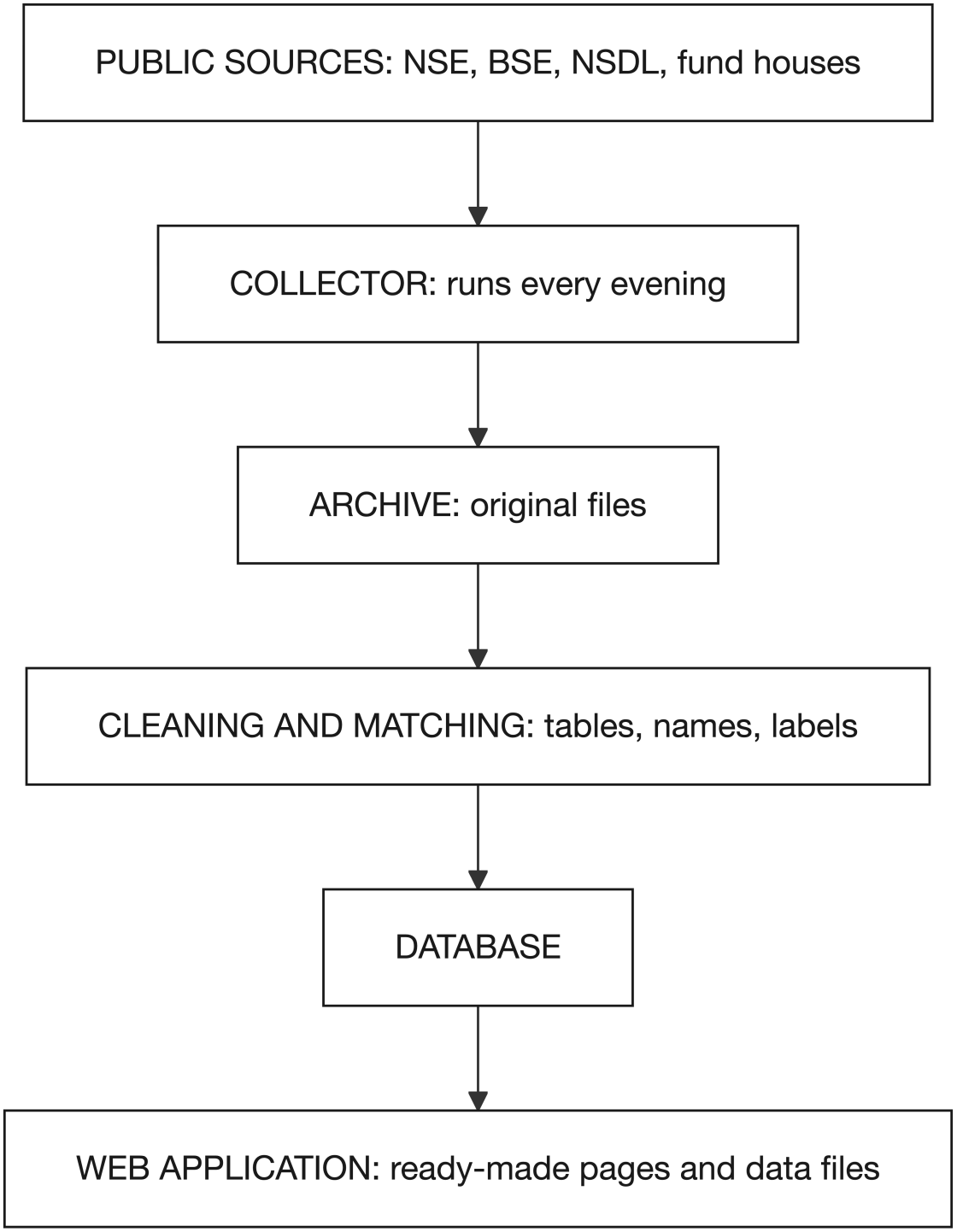
4.6 What is missing

I keep a record of what the data does not have.

Gap	Reason
33 sessions of NSE bulk deals in 2026	NSE no longer serves them
BSE bulk and block deals	The BSE route I tested returned an error page
Some fund houses, such as Kotak, Bandhan and Axis	Their sites block automated access or encrypt the data
Industry labels before 2016	No free source with dates was available
Some large one-day price moves	2,566 are flagged and listed, and 312 of them are not yet explained

5. System architecture

5.1 The big picture



How data travels from a public website to the user.

The design is one straight line. The website never talks to the exchange and never touches the live database. It only reads ready-made files. So the website stays fast, and it keeps working even while new data is being collected.

5.2 What happens every evening

1. At 18:30, the scheduler starts the job.
2. The system downloads the day's files and saves the originals.
3. It reads them into tables, and updates prices, names and labels.
4. It makes an encrypted backup to Google Drive and runs a health check.
5. It rebuilds the website.
6. It sends a report by Telegram and email.

The run has 33 recorded stages. Each stage records whether it passed and how long it took. If one stage fails, the rest continue, and the report names the stage that failed.

A second pass runs at 21:00, for anything published late. A shareholding job runs at 18:45, and a slower backfill job at 21:20.

5.3 Parts of the code

Folder	What it does	Lines of code
`src/archive`	Downloads files and stores the originals	4,872
`src/ingest`	Reads files into tables	4,689
`src/warehouse`	Builds price tables and groups	2,037
`src/identity`	Matches companies and symbols over time	1,345
`src/mart`	Cleans deals and adds flags	700
`src/research`	Measures and studies	8,636
`src/scan`	Wide search over signals	1,978
`src/governance`	Keeps a locked record of experiments	620
`src/monitor`	Health checks, reports and alerts	3,432
`src/site`	Prepares data for the website	2,080
`src/common`	Shared helpers	841
Total		**31,230**

5.4 How the system stays safe

Risk	What protects against it
A file is read wrongly	The original is kept, so it can be read again
The laptop fails	An encrypted backup goes to Google Drive on every run
A night's run fails	An alert goes out by Telegram and email
The website shows old data	Each page shows when it was built, and the System page shows how fresh each source is
A secret key leaks	Keys are stored outside the code, and I run a scan before commits

6. Technology used

Purpose	Tool	Why I chose it
Main language	**Python**	Strong data tools, and easy to test
Analysis database	**DuckDB**	Handles 18 crore rows on a laptop, with no server to run
File storage	**Parquet**	Compressed tables that read very fast
Project records	**SQLite**	Small, and can refuse edits to locked records
Website	**Astro**	Builds fast, simple pages that need no server
Website look	**HTML, CSS, JavaScript**	Interactive charts and tables with few dependencies
Scheduling	**macOS launchd**	Runs jobs by itself and survives restarts
Testing	**pytest**	1,280 automated tests
Version control	**Git and GitHub**	Full history of every change
Backup	**restic and rclone**	Encrypted backups to Google Drive
Alerts	**Telegram and email**	A short report after each run
Report and diagrams	**Markdown, Mermaid, Chrome**	This report is built from text files

Why no server?

A normal website needs a server and a database running all the time. This one does not. All pages and data are prepared in advance, so the website is only a folder of files. That keeps it fast and cheap, and it is simple to look after. This is why one laptop can run the whole project.

Hardware

Everything runs on one laptop with 16 GB of memory. The full test suite alone uses about 8 GB. So heavy jobs are planned so that they do not run at the same time as the evening collection.

AI assistance

I took help from LLM tools by Anthropic for writing code and documents. The project idea, the decisions and the final approvals were mine.

7. The web application

7.1 What it is

The web application is the part of the project people actually see. It brings together everything collected, for 4,419 stocks, 34,601 participants and 3,864 fund schemes. A search box at the top finds any stock, investor or fund.

It runs at `http://127.0.0.1:4321` on my laptop.

7.2 Who uses it now

Right now, I am the only user. The website runs on my laptop and is not published online.

I built it for these people, as future users:

- **Students** learning how Indian markets work
- **Analysts and researchers** who need the data in one place
- **New investors** who want to see the facts before deciding for themselves

Before publishing it, I need to check NSE's rules on sharing its data, because I do not yet have clear permission to redistribute it.

7.3 The pages

Page	What you can do there
Overview	See the latest session, market summary and last collection run
Deals	See every bulk and block deal of any day, with labels
Stocks	See one company's prices, results, owners, deals and filings
Funds	See what each mutual fund scheme holds
Markets	See foreign flows, F&O positions, valuation and trading patterns
Insights	See who trades most and how ownership is changing
Screener	Filter companies by disclosed numbers
Filings	Read announcements and board meetings
Participants	See one investor's trades, holdings and spellings
Watchlist	Save the stocks you follow
Research	Read what the tests found, in plain words
Research lab	Explore how prices moved after events
System	See each run, the data freshness and the backup status
About	Read how each label is decided

7.4 Screens from the application

These were captured from the running application, with data up to 6 November 2026.

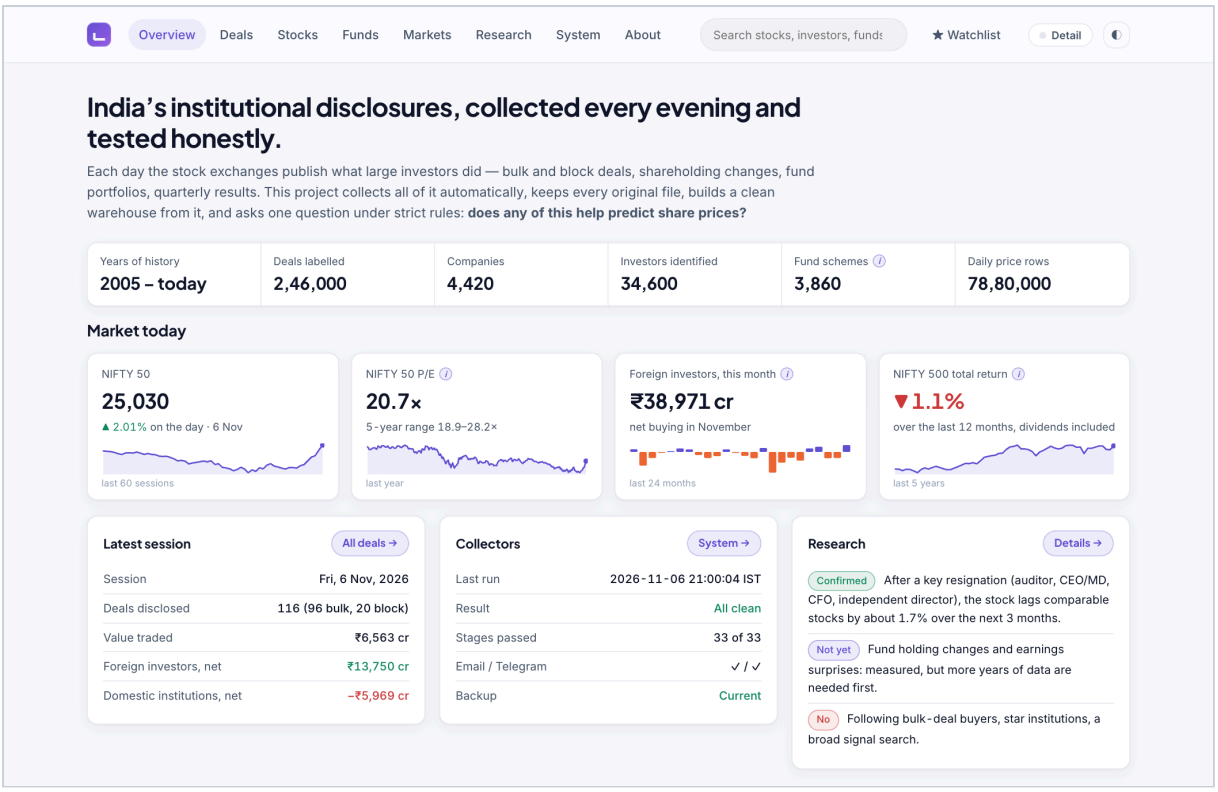


Figure 1. The Overview page: the latest session, the last collection run and the research summary.

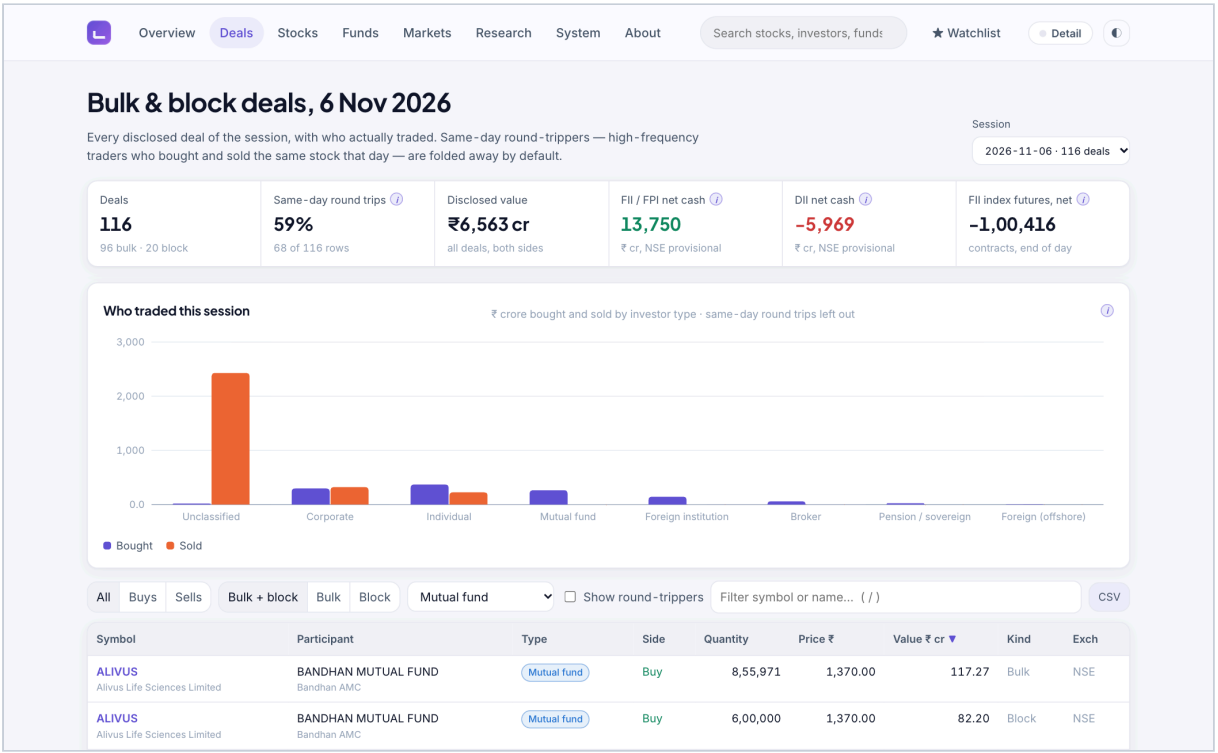


Figure 2. The Deals page, filtered here to mutual funds. Same-day round trips are folded away by default.

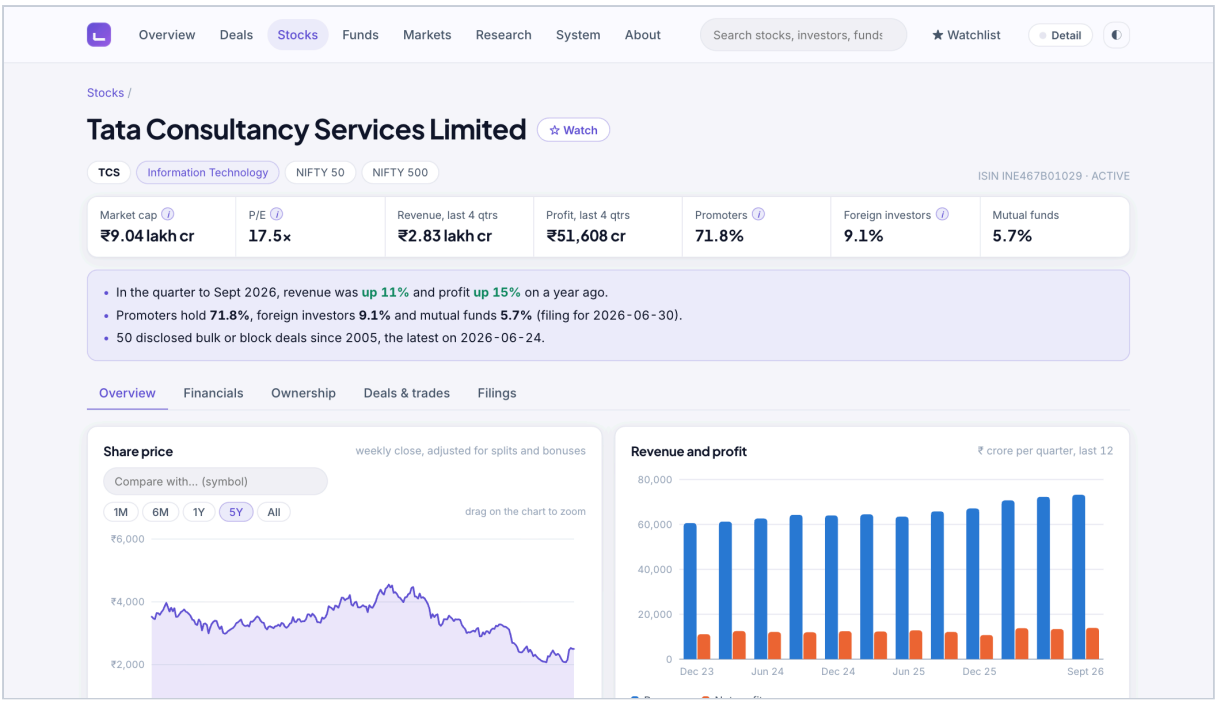


Figure 3. A stock page. All of one company's information is in one place, with no buy or sell label.

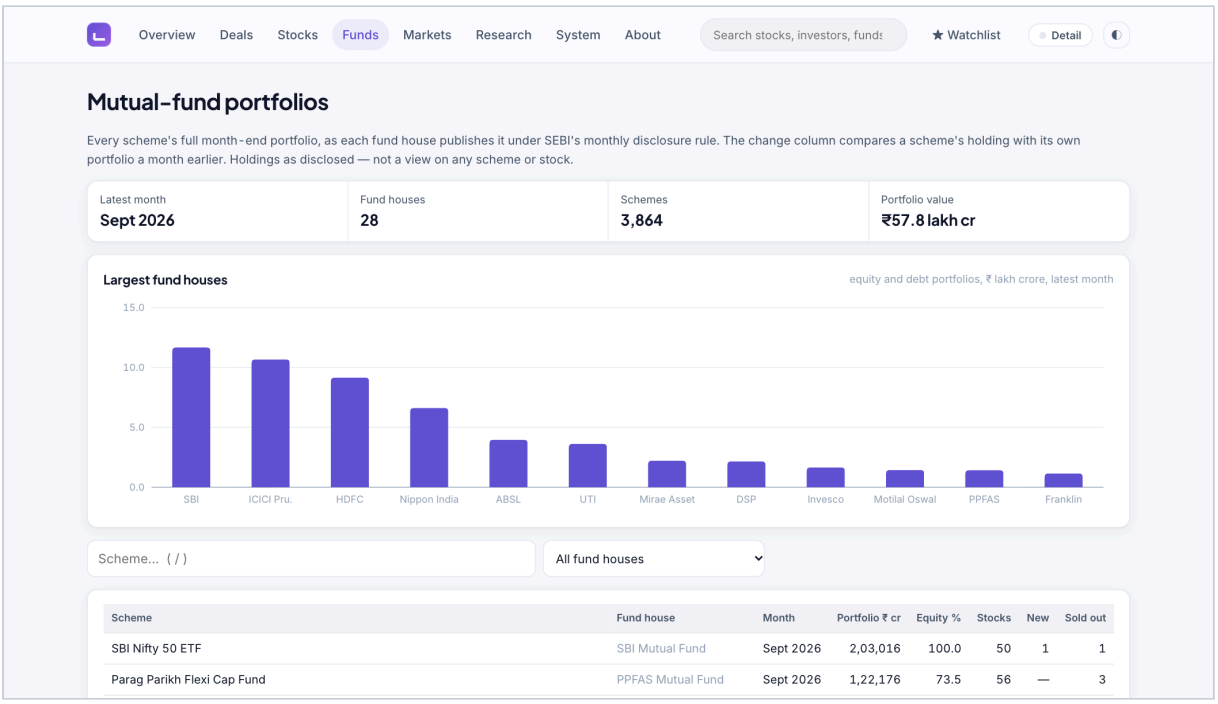


Figure 4. The Funds page: fund houses and their schemes, from month-end portfolios.

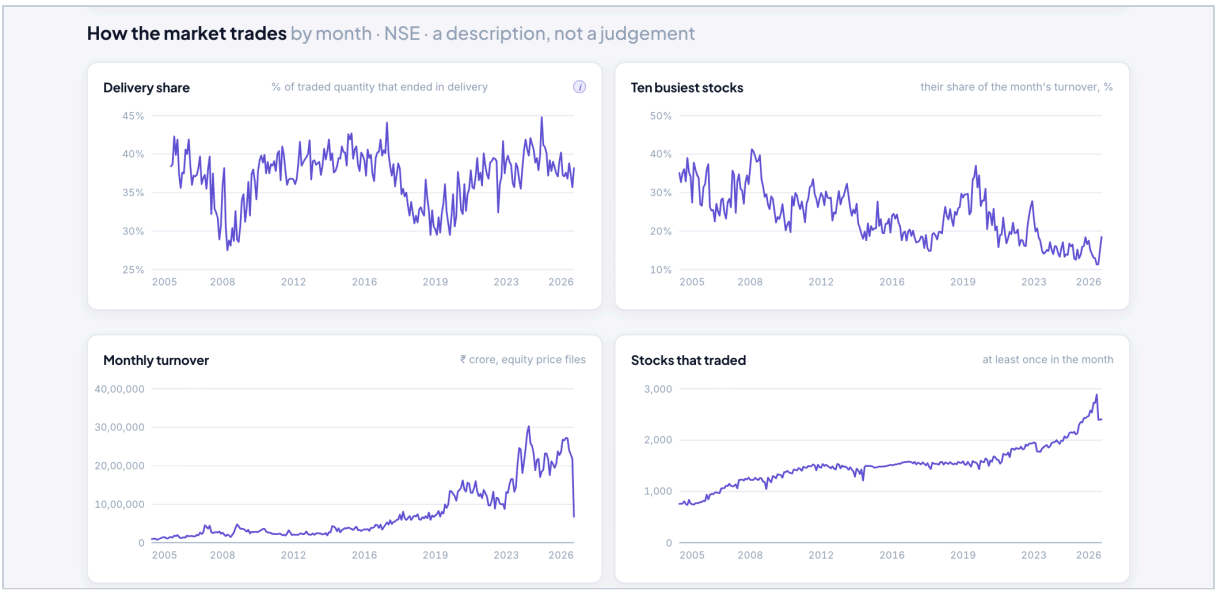


Figure 5. The Markets page, showing how the market trades by month. The fall at the end of the turnover chart is because November 2026 is only part of a month.

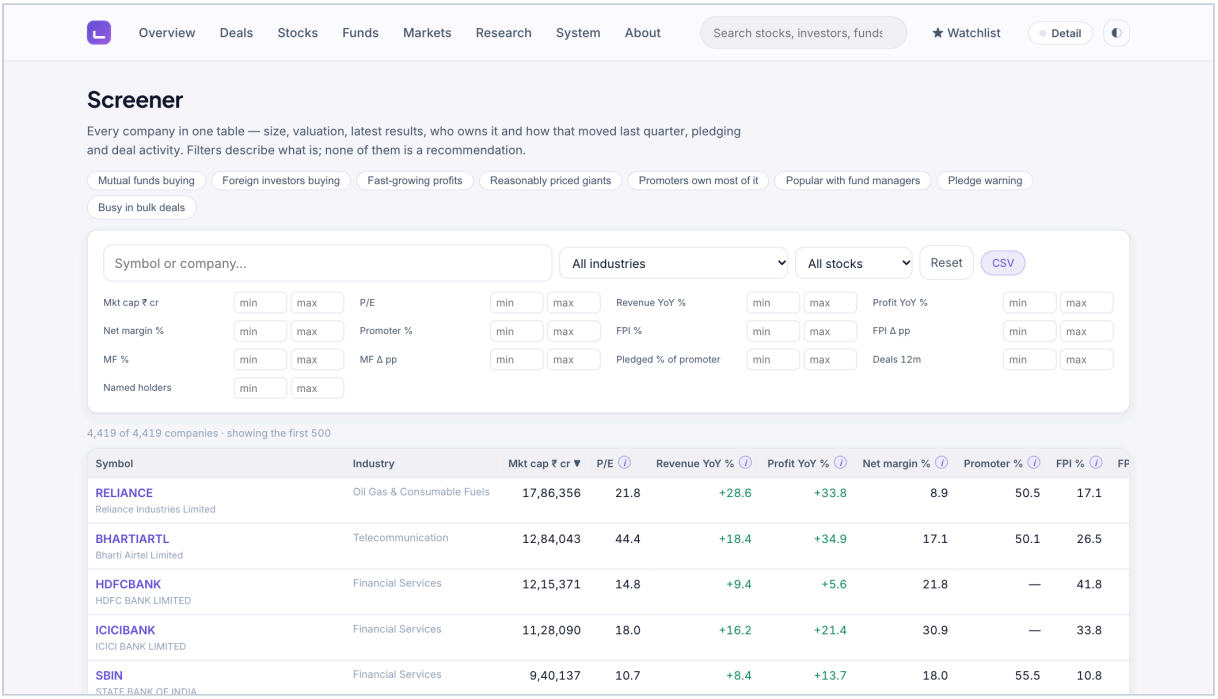


Figure 6. The Screener page: filtering companies on disclosed numbers.

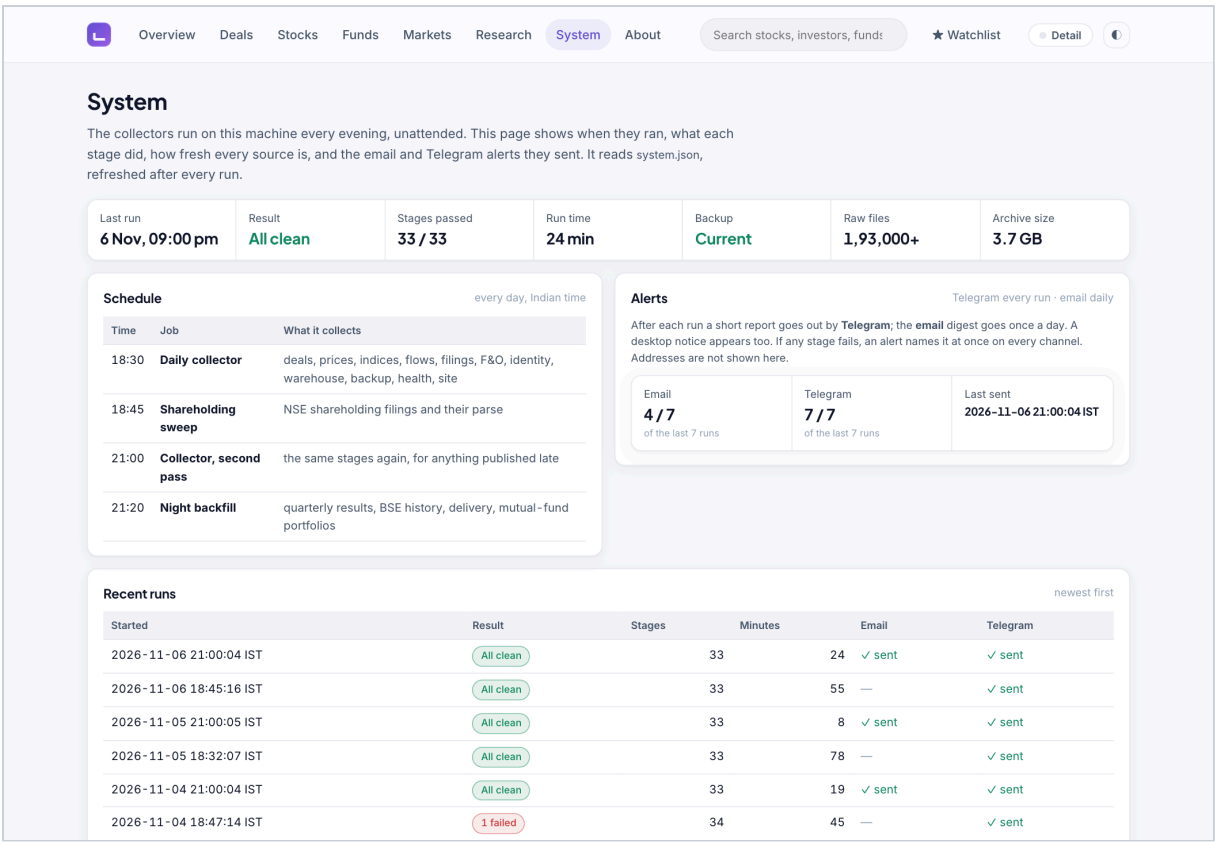


Figure 7. The System page. It shows each run, including a failed one. Nothing is hidden.

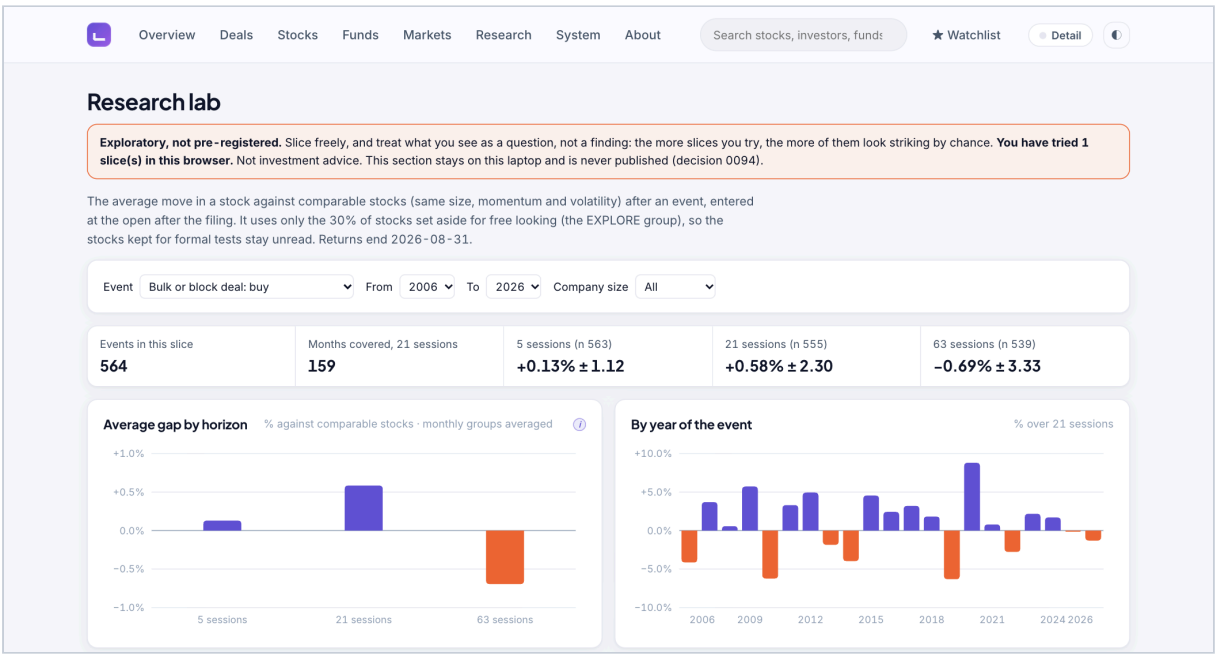


Figure 8. The Research lab, showing one example view. It carries a clear warning, and it stays on my laptop.

7.5 What the data showed

With all this data in one place, I could test some popular beliefs. For each test, I wrote down the question and the pass mark before looking at the answer, so I could not change them afterwards.

Question	Answer
Does avoiding shares after a bulk deal improve a portfolio?	No
Do some institutions keep picking winners?	No
Do F&O positions predict the market?	Cannot say: the data is about four times too short
Does a rise in institutional holding predict the share?	Cannot say: the data is too short
Can a very wide computer search find a working signal?	No
Does a stock fall behind similar stocks after a key person resigns?	**Yes**

The one "yes" is this: after a company reports that its auditor, CEO or MD, CFO or an independent director has resigned, its share falls behind similar shares by about 1.7% on average over the next three months. This is based on 4,894 cases.

This is not a way to make money. To use it, you would have to sell shares short, and the costs of doing that were not included. It only describes how prices behaved.

The result that matters most for this project is that public disclosures are useful to read, but they are not a simple signal to buy or sell. That is one more reason the website only shows facts.

Figure 9 shows these answers on the website. The Research lab in Figure 8 lets me explore this kind of question further, but its numbers are only for exploring and are not findings.

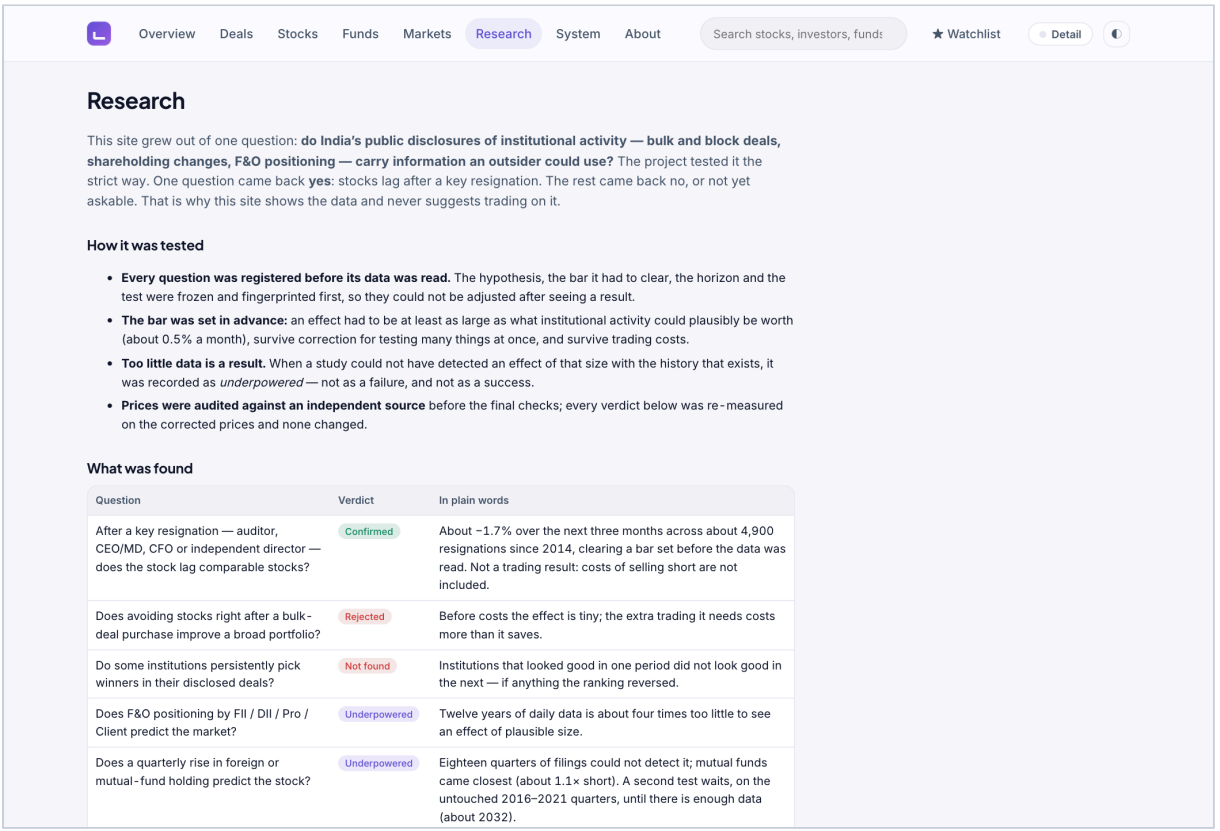


Figure 9. The Research page: what the tests found, in plain language.

8. GitHub status

8.1 Where the code is

Repository	`https://github.com/markandeyavarma3-lab/MICC_V3`
Visibility	Public
Main branch	`main`
Commits	313, since 16 August 2026
Automated tests	**1280 automated tests** in 103 files
Automatic checks	GitHub Actions runs the basic tests on each push. The latest run, on 7 November 2026, passed

The website has its own separate repository, `research-website`, with 26 commits. That one stays only on my laptop, because the website is not published.

8.2 What is on GitHub and what is not

The source code, tests, plans, decision records and reports are on GitHub. The data (about 19 GB), the databases, and all passwords and keys are not. I run a secret scan before commits, because the repository is public.

8.3 Work over time

The repository received 54 commits in August 2026, 116 in September, 128 in October and 15 so far in November.

8.4 Latest commits

Commit	Date	What changed
`aff8e78`	2026-11-07	CI: run on macOS, where the project runs
`a700404`	2026-11-07	CI: install zsh, which the backup tests execute
`209de3b`	2026-11-07	CI: failing test names as public annotations
`87a3c4a`	2026-11-07	Review report: nine plain-English parts, Arial, with screens and the commit log
`97ee4db`	2026-11-07	Research lab, market trading patterns, and GitHub's unit tests made green
`47a9c8c`	2026-11-06	Shareholding through renames and typos; mis-scaled results withheld; BSE filings since 2021
`57eaecf`	2026-11-06	Site: SME prices from the bhavcopies, as traded
`77dd67c`	2026-11-06	Site: investor labels corrected from the exchange's raw names; research untouched

8.5 Earlier versions

This is the third version of the project. V1 (June to July 2026) collected the first data, 1.13 crore rows that are still in use. V2 (July to August 2026) tried to find trading strategies, and I stopped it after an audit found its costs were calculated wrongly.

9. Conclusion

What was achieved

India's free market disclosures were scattered, some disappeared daily, and they came in many formats with many names for the same thing. This project brings them together.

- **The collector** runs by itself every evening, and has saved more than 1,92,000 original files.
- **The database** holds about 20 years of market history, with companies and investors matched across name changes.
- **The web application** shows 4,419 stocks, 34,601 participants and 3,864 fund schemes in one place, with no advice.
- **The checks** include 1,280 automated tests, a price check against a separate source, and a health report after every run.

What is not finished

- It runs on one laptop. If the laptop is off, data that is only available for a day can be lost.
- The website is not public yet, as NSE's data rules still have to be checked.
- Some data is missing, as listed in Section 4.6.
- The same trade can appear in both the bulk and block lists, so a day's total value on the website may count it twice.

What I learned

- **Collecting data is harder than analysing it.** Most of my time went into getting clean, correct data.
- **Measure, do not assume.** Several times, my own earlier results turned out to be wrong when I checked them properly.
- **Save the original.** Every time I found a mistake, the saved file let me fix it.
- **A "no" is a real answer.** Most of the ideas I tested did not work, and knowing this is useful.

What comes next

- Check NSE's data rules, then decide about publishing the website.
- Keep collecting, so that the questions which are short of data today can be answered later.
- Add the fund houses and sources that are still missing, where they allow it.

Final word

Free data is only useful if people can find it, trust it and understand it. This project takes India's public market disclosures, which were spread across many places, and makes them available in one place that is simple to read.